

Course Project Report

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Project 1: Acoustic Link

Introduction

Project 1 introduces the fundamental concepts of utilizing audio interfaces to establish a data link between two air-gapped computer nodes. The focus of this project is to manipulate and detect sound signals to facilitate communication between nodes. Specifically, NODE1 will generate audio signals, which will be transmitted over an acoustic channel, and NODE2 will receive and decode the transmitted information. This project emphasizes key physical-layer protocol elements, including modulation methods, frame structures, error correction, and transmission/reception interfaces.

The tasks within this project are designed to explore and implement the following:

- Modulation methods for encoding data onto audio signals.
- Frame structure design for transmitting information reliably.
- Error correction mechanisms to ensure data integrity despite potential interference or noise.
- Creating both Tx (transmit) and Rx (receive) interfaces to interact with upper-layer protocols.

Task 1: Understanding Your Tools

Follow the instructions in the documentation to set up the device.

Task 2: Generating Sound Waves at Will

The first task involves NODE1 generating a complex audio signal composed of two sinusoidal components: a low-frequency tone at 1 kHz and a high-frequency tone at 10 kHz. The signal is defined as: $f(t) = \sin(2\pi \cdot 1000 \cdot t) + \sin(2\pi \cdot 10000 \cdot t)$.

This is a task helping us understanding the audio API so there is no need to talk much about it.

Task 3: Transmitting First Bit

Objective

In this task, NODE1 will transmit data through the audio network to NODE2. Specifically, the goal is to send 10,000 bits, encoded as **0** or **1**, using the generated audio signal. This task demonstrates the ability to transmit information using modulated sound.

Implementing Steps

- Generating a preamble. We chose the Chrip signal, whose frequency decreases with time.
- Do the modulation. As we are transmitting the binary data, encoding is not necessary. Here we use a sinusoidal signal as the carrier wave and use phase shift to modulate.
- Synchronize the preamble. We implemented a sliding window here that will process the dot product with a standard preamble when the power goes to a peak and the synchronization process is finished.
- Demodulation. After receiving the sound track, demodulate it back to binary format.

Task 4: Error Correction

Objective

Due to the unstableness of the sound transmission, errors may occur during the transmission. We can use some method to detect and correct the error.

Method

In this task we use Hamming codes to correct the error. Hamming codes can detect one-bit and two-bit errors, or correct one-bit errors without detection of uncorrected errors. By contrast, the simple parity code cannot correct errors, and can detect only an odd number of bits in error.

Task 8: Range Challenge

In this task we need to achieve longer data transmission distances. We keep the previous transmission method and still use Hamming codes to correct errors, but we use more advanced loud-speaker and microphone to improve the transmission quality. We succeed in a 150 centimeter transmission distance at last.

Challenges

Transmitting data through sound wave in the air would be greatly affected by the environment, so we have struggled to adjust the arguments in our code, like the frequency of the carrier wave, by observing the image of the recorded sound wave to achieve the best result.

Project 2: Manage Multiple Access

Introduction

Building upon the concepts introduced in Project 1, Project 2 aims to extend reliable data transmission to multiple devices. Rather than relying on sound signals, this project leverages audio cables to transmit analog electrical signals. These cables drive the loudspeakers and effectively serve as an “electromagnetic mirror” of the sound, offering improved transmission quality. With the use of wired connections, this approach eliminates many of the environmental factors (such as noise and signal degradation) associated with sound waves. The physical layer is thus simplified, and transmission becomes more stable and faster.

Task 1: Cable Connection

Objective

Use cable connection to transfer electrical signal directly, to form a faster and more stable physical layer.

Implementing Details

For a higher performance, we change the modulation method from use sinusoidal signal to square wave, which performs better when transmitting, however, the disadvantage is that it cannot handle a sequence of all 0 or 1. However, due to the particularity of the task, we assume that the binaries are randomly generated. If we really want to transmit a real TCP packet in later session, we can use a map from 5 Bit to 4 Bit to avoid continuous 0 or 1.

Task 2: Acknowledgement

Objective

Despite the advantages of using wired connections, errors and collisions can still occur, especially when multiple devices share the same cable. Task 2 aims to address these issues by implementing a mechanism to increase the transmission reliability, focusing on collision detection and error acknowledgment (ACK).

Method

In this task we implement stop-and-wait, which means after transmitting a single frame, we will not transmit the next one until we receive the ACK.

Implementing Steps

- Transmitter begins to prepare frames.
- Once a frame is ready, then send this frame and waiting for ACK. At the same time, setup a ACK timer.
- Once the receiver heard the frame, it checks the correctness and send ACK to receiver.

- If the transmitter receives ACK, which means the past frame has successfully transmitted, it begins to send the next ready frame.
- If ACK loss, the transmitter should start a retransmission.

Challenges

We once tried to complete task 6, which is transmitting data simultaneously, by enabling 4 nodes to take turns transmitting one frame so that carrier sense might be ignored, but the frame of one node tended to be partly lost during our test.

Project 3: To The Internet

Introduction

In this project, the goal is to interconnect the Aethernet with operational networks, leveraging the Internet Protocol (IP) as defined in RFC 791. IP is integral to the Internet, enabling the interconnection of diverse networks. As illustrated in Figure 5.1, IP nodes are divided into two categories:

- **Router Nodes:** Equipped with multiple network interfaces, routers handle packet forwarding and network interconnection.
- **Host Nodes:** All non-router nodes in the network, which use IP for communication within and across networks.

Task 1: ICMP Echo

Objective

Internet Control Message Protocol (ICMP) Internet control Message Protocol. It is a subprotocol of the TCP/IP protocol family and is used to pass control messages between IP hosts and routers. In this task, we use it to verify the reachability from the host to the router. The Aethernet host will try to ping the router, and the later should be able to send echo reply.

Implementing Details

Since the physical layer has already been set up in the previous task, we will only focus on the ICMP packet construction process. ICMP is part of the Internet protocol suite as defined in RFC 792, so we will generate packet according to this standard.

- **ICMP echo request:** The ICMP header starts after the IPv4 header and is identified by its protocol number, All ICMP packets have an eight-byte header and variable-sized data section. For echo request, the field **type** is 8 and the field **code** is 0. The following sections are the checksum, identifier, and sequence number.

- **ICMP echo reply:** Very easy to generate, just change the **type** field to 0 and recalculate the checksum.

For simplicity, we don't generate an actual ICMP echo request package here, but this knowledge will be used in later tasks.

Task 2: Router

Objective

In this task, our Aethernet should be able to respond to ping requests from other networks, and this property needs to be implemented through the router.

Implementing Details

we use another device to send an icmp echo request over a hotspot connection to the router, which needs to receive the packet and forward it to the destination (in this case, a host in Aethernet), and then reply to the device via the hotspot after receiving the echo reply. Here we use pcap library to capture the packet.

Task 3: NAT

Objective

Network Address Translation (NAT) refers to network address translation, proposed in 1994. NAT is a technique for using private addresses on the local network and switching to global IP addresses when connecting to the Internet. In this task we should change the IP address in the packet which from Aethernet to public address.

Implementing Details

If an Intranet address is used to transmit an echo request, the echo reply cannot be received. Therefore, a router is required to translate the packet. In this case, we directly use the IP address of the router in the campus network.

In the concrete implementation, when the device connected by the hotspot executes the ping command, its packet will be automatically forwarded by computer, so we only need to implement the part of Aethernet forwarding to the campus network.

Project 4: Above IP

Introduction

In previous projects, we have implemented a basic network using audio lines, and in this chapter, we will implement something else related to it.

Task 1: DNS

Objective

The Domain Name System (DNS) is a hierarchical and distributed name service that provides a naming system for computers, services, and other resources on the Internet or other Internet Protocol (IP) networks. In this task we need to configure the router from the previous task to be the local DNS server of the Aethernet host.

Implementing Details

The Aethernet host will still use the **ping** command, but the argument will be the domain name instead of the IP address. Router should be able to resolve the domain name and reply to the host. In the actual implementation, we took a slightly lazy approach and used the boost library to query domain names, so we only needed to make some corresponding interfaces.

More thoughts on this course

During the process of completing the course project, we have encountered multiple issues that required debugging hardware devices. In fact, during project 0, which was the preparatory stage of the entire project, we found that the sound-card driver was incompatible with one of the laptops and eventually had to switch to another laptop to complete the environment setup. In project 1, we were also repeatedly troubled by the transmission quality of sound through the air. Having advanced sound speakers and microphones could better solve this problem. It would be great if the requirements of the devices could be better handled.

The projects in this course are highly demanding of time and engineering skills, and those who complete them will gain a great deal. We really appreciate the professor's guidance and the teaching assistants' support. Wish CS120 continues to improve and thrive!